16 An ideal gas is held in a container of volume V at pressure P. The root-mean-square (r.m.s) speed of a gas molecule under these conditions is u. If the volume and pressure are now changed to $2V$ and $2P$ respectively, what will be the r.m.s speed of a molecule?

A u **B** $\sqrt{2}u$ **C** $2u$ **D** $4u$

Ans: C

Mean kinetic energy, $KE = \frac{3}{2}kT$ and Ideal gas equation, $pV = NkT$

Therefore, we have $KE = \frac{3}{2}\frac{pV}{N} \Rightarrow \frac{1}{2}m\langle c^2 \rangle = \frac{3}{2}\frac{pV}{N}$

$$\Rightarrow c_{rms} = \sqrt{\frac{3pV}{Nm}}$$

$$\Rightarrow c_{rms} \propto \sqrt{pV}$$

$$\Rightarrow \frac{u'}{u} = \sqrt{\frac{(2P)(2V)}{PV}} = 2$$

Required rms speed of the molecule, $u' = 2u$